

## Question Bank

### Unit IV: Introduction to Robotics (12 Hrs.)

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**Syllabus:** Introduction to robotics: synthesis of elements with movability constraints; elements of robot anatomy; hydraulic, pneumatic and electrical manipulators; end-effectors and their design; controllers with microprocessors or fluidics; robot sensors; applications of industrial robots; economics of robotics.

**Part A** 10 multiple-choice questions  $10 \times 1 = 10$  marks

**Part B** 5 short-answer questions  $5 \times 2 = 10$  marks

**Part C** 3 long-answer questions  $3 \times 10 = 30$  marks

#### Part A: Multiple-Choice Questions

1 mark each

**Q1.** A robot whose arm has three linear (sliding) joints is called

- (a) Cylindrical
- (b) Cartesian
- (c) Polar
- (d) Articulated

**Answer:** (b) Cartesian (gantry) — LOO joints, box-shaped work envelope.

**Q2.** SCARA stands for

- (a) Simple Compliant Automatic Robot Arm
- (b) Sensor Controlled Assembly Robot Arm
- (c) Serial Controlled Articulated Robot Arm
- (d) Selective Compliance Assembly Robot Arm

**Answer:** (d) Selective Compliance Assembly Robot Arm.

**Q3.** A free rigid body in space has how many degrees of freedom?

- (a) 3
- (b) 4
- (c) 6
- (d) 12

**Answer:** (c) 6 — three translations and three rotations.

**Q4.** A revolute (R) joint provides how many degrees of freedom?

- (a) 1
- (b) 0
- (c) 2
- (d) 3

**Answer:** (a) 1 — rotation about one axis.

**Q5.** A gripper that lifts flat glass sheets by suction is a

- (a) magnetic gripper
- (b) hook
- (c) mechanical gripper
- (d) vacuum gripper

**Answer:** (d) Vacuum gripper (suction cups).

**Q6.** The sensor most commonly used to measure robot joint position is the

- (a) thermocouple
- (b) encoder
- (c) strain gauge
- (d) microphone

**Answer:** (b) Encoder (incremental or absolute).

**Q7.** The first industrial robot, installed at General Motors in 1961, was

- (a) PUMA (b) SCARA  
(c) Unimate (d) ASIMO

**Answer:** (c) Unimate (hydraulically driven).

**Q8.** Fluidic logic controllers use

- (a) jets of air or liquid (b) transistors  
(c) optical fibres (d) relays

**Answer:** (a) Jets of air or liquid, with no moving parts (Coanda / wall-attachment effect).

**Q9.** Which drive system gives the highest payload capacity?

- (a) Pneumatic (b) Solenoid  
(c) Stepper motor (d) Hydraulic

**Answer:** (d) Hydraulic.

**Q10.** The work envelope of a cylindrical robot is shaped like a

- (a) box (b) hollow cylinder  
(c) sphere (d) cone

**Answer:** (b) Hollow cylinder (portion of an annular cylinder).

### Part B: Short-Answer Questions

2 marks each

**Q1.** State the Grübler–Kutzbach mobility equation for planar mechanisms and find the mobility of a four-bar linkage. [2]

#### Answer

$M = 3(n - 1) - 2j_1 - j_2$ , where  $n$  = number of links (including the fixed link),  $j_1$  = 1-DOF joints,  $j_2$  = 2-DOF joints.

Four-bar:  $n = 4$ ,  $j_1 = 4 \Rightarrow M = 3(3) - 2(4) = 1$  — one input (motor) drives it.

**Q2.** Differentiate between accuracy and repeatability of a robot. [2]

#### Answer

**Accuracy** is how closely the robot reaches a *commanded* point in space: accuracy =  $CR/2 + 3\sigma$ . **Repeatability** is how closely it returns to a previously *taught* point:  $\pm 3\sigma$ . Robots are usually much more repeatable than accurate.

**Q3.** A 2 kg part is held between two fingers with coefficient of friction 0.3. Find the gripping force needed to hold it at rest (take  $g = 9.81 \text{ m/s}^2$ ). [2]

#### Answer

$$\mu n_f F_g = mg \Rightarrow F_g = \frac{mg}{\mu n_f} = \frac{2 \times 9.81}{0.3 \times 2} = 32.7 \text{ N (a safety factor, e.g. 2, is then applied).}$$

**Q4.** Distinguish between internal and external robot sensors, with two examples of each. [2]

**Answer**

**Internal sensors** measure the robot's own state — e.g. encoders (joint position), tachogenerators (joint velocity). **External sensors** measure the environment — e.g. proximity sensors, force–torque sensors, vision cameras, touch sensors.

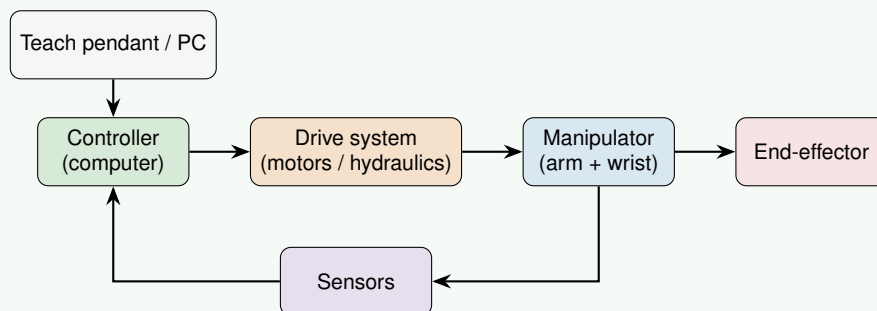
- Q5.** A robot installation costs Rs 40 lakh and gives a net saving of Rs 12 lakh per year. Find the payback period and the return on investment. [2]

**Answer**

Payback =  $40/12 \approx 3.3$  years; ROI =  $12/40 \times 100 = 30\%$  per year.

**Part C: Long-Answer Questions****10 marks each**

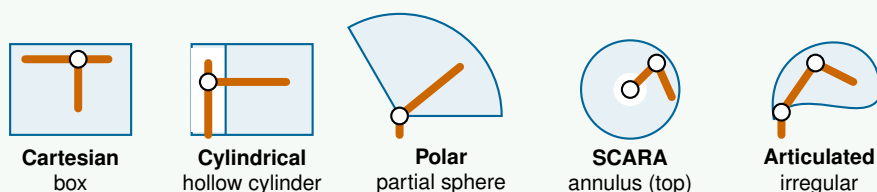
- Q1.** Explain the components of an industrial robot system with a block diagram. Describe the elements of robot anatomy and the five common robot configurations with their work envelopes and applications. [10]

**Answer****(a) Robot system**

The **controller** stores the program and plans the motion; the **drive system** powers the joints; the **manipulator** moves in space; the **end-effector** does the job; **sensors** feed back the arm position and information about the surroundings.

**(b) Robot anatomy**

- **Base** — fixed to floor, ceiling or a track.
- **Arm (body)** — first three joints (waist, shoulder, elbow); **position** the wrist in space.
- **Wrist** — last 2–3 joints (roll, pitch, yaw); **orient** the tool.
- **End-effector** — gripper or tool on the wrist flange; the **tool centre point** is the point that is programmed.
- Joints: linear (L), orthogonal (O) — prismatic; rotational (R), twisting (T), revolving (V) — revolute.

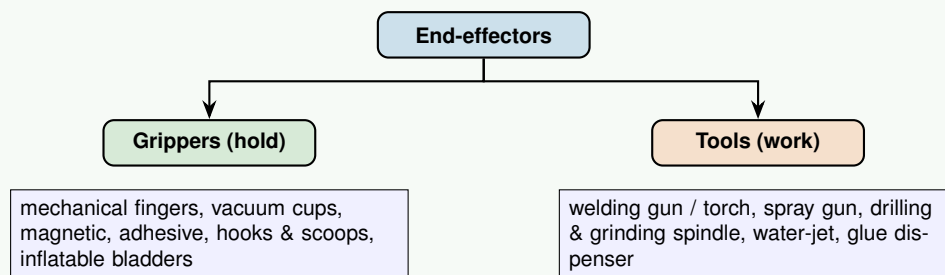
**(c) Configurations**

| Configuration | Joints | Feature                | Application                 |
|---------------|--------|------------------------|-----------------------------|
| Cartesian     | LOO    | rigid, accurate        | gantry loading, palletising |
| Cylindrical   | TLO    | large vertical reach   | machine loading             |
| Polar         | TRL    | long horizontal reach  | spot welding, die casting   |
| SCARA         | VRO    | fast, stiff vertically | electronic assembly         |
| Articulated   | TRR    | dexterous, compact     | arc welding, painting       |

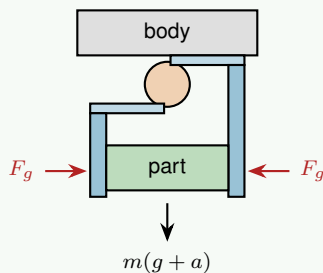
- Q2.** Classify end-effectors with a neat diagram. Explain the working of a mechanical gripper and derive the expression for the gripping force. Design a vacuum cup to lift a 5 kg plate at 60 kPa vacuum with a safety factor of 2. [10]

### Answer

#### (a) Classification



#### (b) Mechanical gripper



A pneumatic cylinder or motor turns a pinion that moves two racks in opposite directions, so both fingers close equally on the part (rack-and-pinion parallel gripper). Other mechanisms: linkages (toggle), cams, screws.

**Gripping force:** the friction at all contact surfaces must support the weight and inertia force:

$$\mu n_f F_g \geq m(g + a) \Rightarrow F_g = \frac{m(g + a)}{\mu n_f} \times \text{SF}$$

$\mu$  = friction coefficient,  $n_f$  = number of contact surfaces,  $a$  = acceleration, SF = safety factor.

e.g.  $m = 2 \text{ kg}$ ,  $\mu = 0.3$ ,  $n_f = 2$ ,  $a = 10 \text{ m/s}^2$ , SF = 2:  $F_g = \frac{2(19.81)}{0.6} \times 2 \approx 132 \text{ N}$ .

#### (c) Vacuum cup design ( $m = 5 \text{ kg}$ , $\Delta p = 60 \text{ kPa}$ , SF = 2)

$$A = \frac{\text{SF} \cdot mg}{\Delta p} = \frac{2 \times 5 \times 9.81}{60\,000} = 1.64 \times 10^{-3} \text{ m}^2, \quad D = \sqrt{\frac{4A}{\pi}} \approx 46 \text{ mm}$$

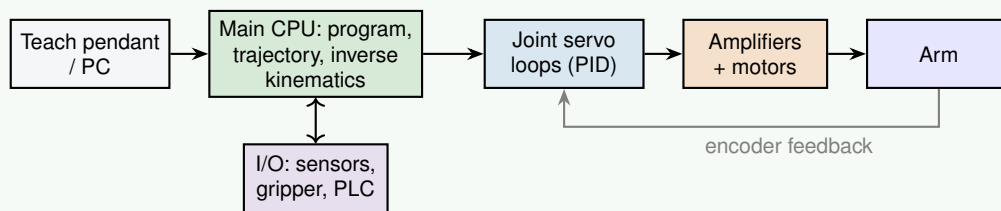
A standard 50 mm cup would be chosen.

**(d) Design considerations:** part mass, shape, surface and fragility; gripping method; end-effector weight (reduces payload); tolerance and need for compliance (RCC device); sensing (part-present switch); power supply (air, vacuum, electric); quick tool change.

- Q3.** Explain the levels of robot control. With a block diagram, explain a microprocessor-based robot controller, and describe fluidic control. Compare microprocessor and fluidic controllers. [10]

**Answer****(a) Levels of robot control**

- **Limited-sequence (bang-bang)** — joints move between mechanical stops; sequence from a timer or logic (fluidics, PLC, cams).
- **Playback, point-to-point** — taught points stored; the path between them is not controlled.
- **Playback, continuous path** — the whole path (lines, arcs) is controlled by interpolation.
- **Intelligent** — uses sensors and vision to change its actions; takes decisions.

**(b) Microprocessor-based controller**

The **main processor** interprets the program, plans the path and solves the inverse kinematics every few milliseconds. **Joint processors** run a PID position loop for each joint at 1–10 kHz using encoder feedback. The **I/O** section handles grippers, conveyors, machine tools and safety interlocks. Programs are entered by teach pendant, lead-through or off-line programming.

**(c) Fluidic control:** logic elements made of channels in which a **jet of air** attaches to one wall (Coanda effect); a small control jet switches it to the other outlet — a binary element with **no moving parts**. AND, OR, NOT, NOR, flip-flops and timers can be built and connected directly to pneumatic cylinders, making simple **limited-sequence pick-and-place robots**.

**(d) Comparison**

| Feature     | Microprocessor                             | Fluidic                                       |
|-------------|--------------------------------------------|-----------------------------------------------|
| Signal      | electrical bits                            | air-jet pressure (0/1)                        |
| Speed       | microseconds                               | milliseconds                                  |
| Functions   | arithmetic, kinematics, PID, vision        | ON/OFF logic and sequencing only              |
| Programming | software, easy to change                   | re-plumb the circuit                          |
| Environment | needs protection from heat, EMI, sparks    | safe in explosive, hot, radioactive areas     |
| Typical use | servo, continuous-path, intelligent robots | limited-sequence robots      pneumatic robots |